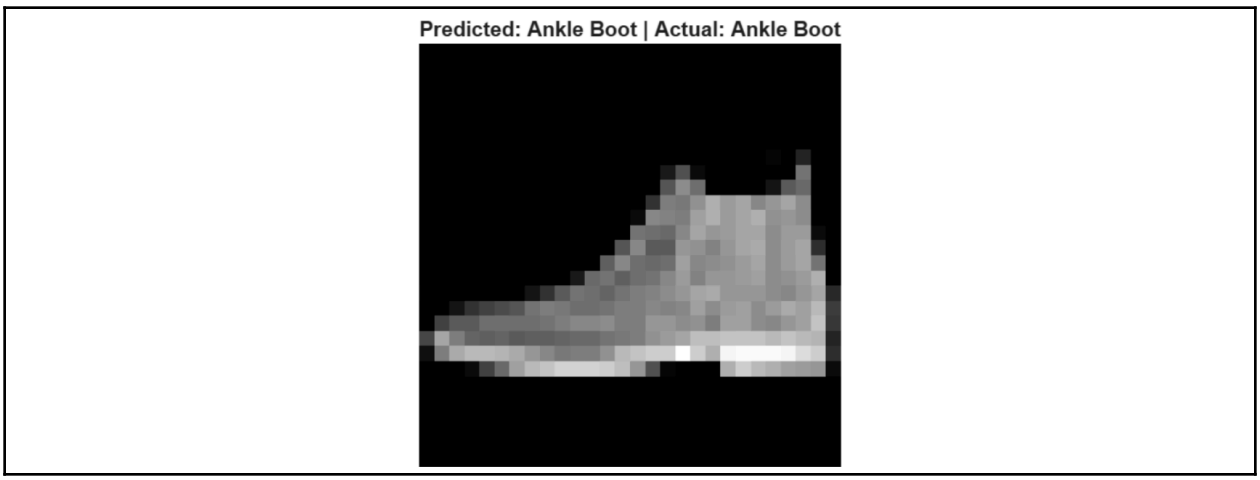
## Experiment 8

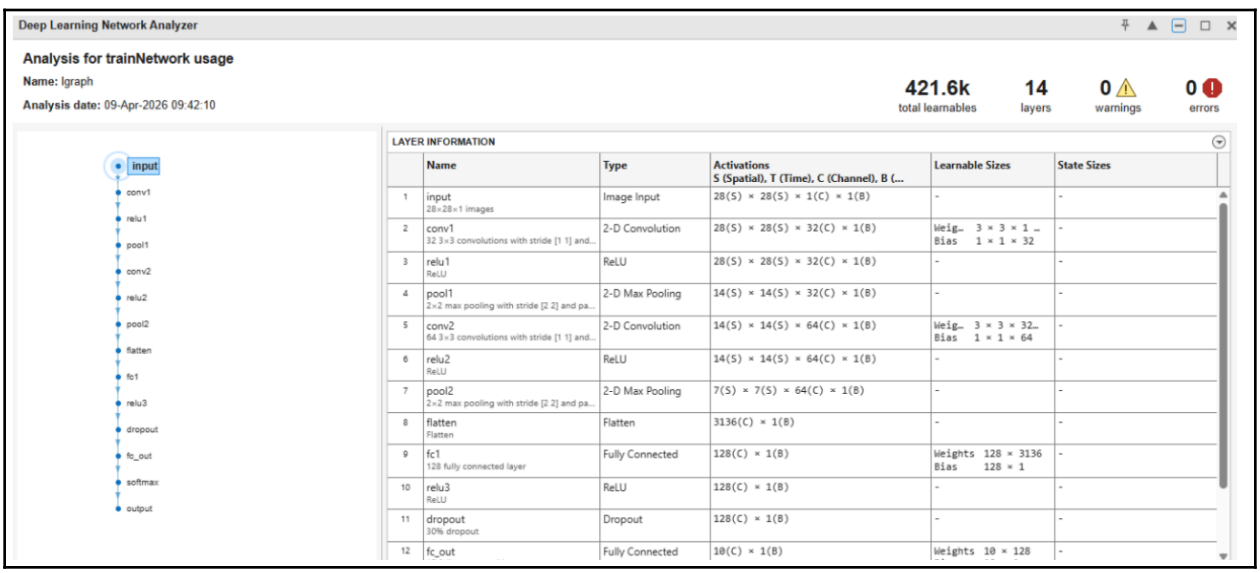
**Aim :**Design and train a Convolutional Neural Network (CNN) on image datasets (e.g., MNIST/CIFAR)

**Data Source :**

- **Total images:** 70,000
- **Training:** 60,000
- **Testing:** 10,000
- **Image size:** 28×28 pixels
- **Format:** Grayscale
- **Classes:** 10 (clothing items)
- [Dataset Link]

**Output :**




| | Name | Type | Activations S (Spatial), T (Time), C (Channel), B (... | Learnable Sizes | State Sizes |
|---|---|---|---|---|---|
| 1 | input 28×28×1 images | Image Input | 28(S) × 28(S) × 1(C) × 1(B) | - | - |
| 2 | conv1 32 3×3 convolutions with stride [1 1] and... | 2-D Convolution | 28(S) × 28(S) × 32(C) × 1(B) | Weig... 3 × 3 × 1 ... Bias 1 × 1 × 32 | - |
| 3 | relu1 ReLU | ReLU | 28(S) × 28(S) × 32(C) × 1(B) | - | - |
| 4 | pool1 2×2 max pooling with stride [2 2] and pa... | 2-D Max Pooling | 14(S) × 14(S) × 32(C) × 1(B) | - | - |
| 5 | conv2 64 3×3 convolutions with stride [1 1] and... | 2-D Convolution | 14(S) × 14(S) × 64(C) × 1(B) | Weig... 3 × 3 × 32... Bias 1 × 1 × 64 | - |
| 6 | relu2 ReLU | ReLU | 14(S) × 14(S) × 64(C) × 1(B) | - | - |
| 7 | pool2 2×2 max pooling with stride [2 2] and pa... | 2-D Max Pooling | 7(S) × 7(S) × 64(C) × 1(B) | - | - |
| 8 | flatten Flatten | Flatten | 3136(C) × 1(B) | - | - |
| 9 | fc1 128 fully connected layer | Fully Connected | 128(C) × 1(B) | Weights 128 × 3136 Bias 128 × 1 | - |
| 10 | relu3 ReLU | ReLU | 128(C) × 1(B) | - | - |
| 11 | dropout 30% dropout | Dropout | 128(C) × 1(B) | - | - |
| 12 | fc_out | Fully Connected | 10(C) × 1(B) | Weights 10 × 128 | - |

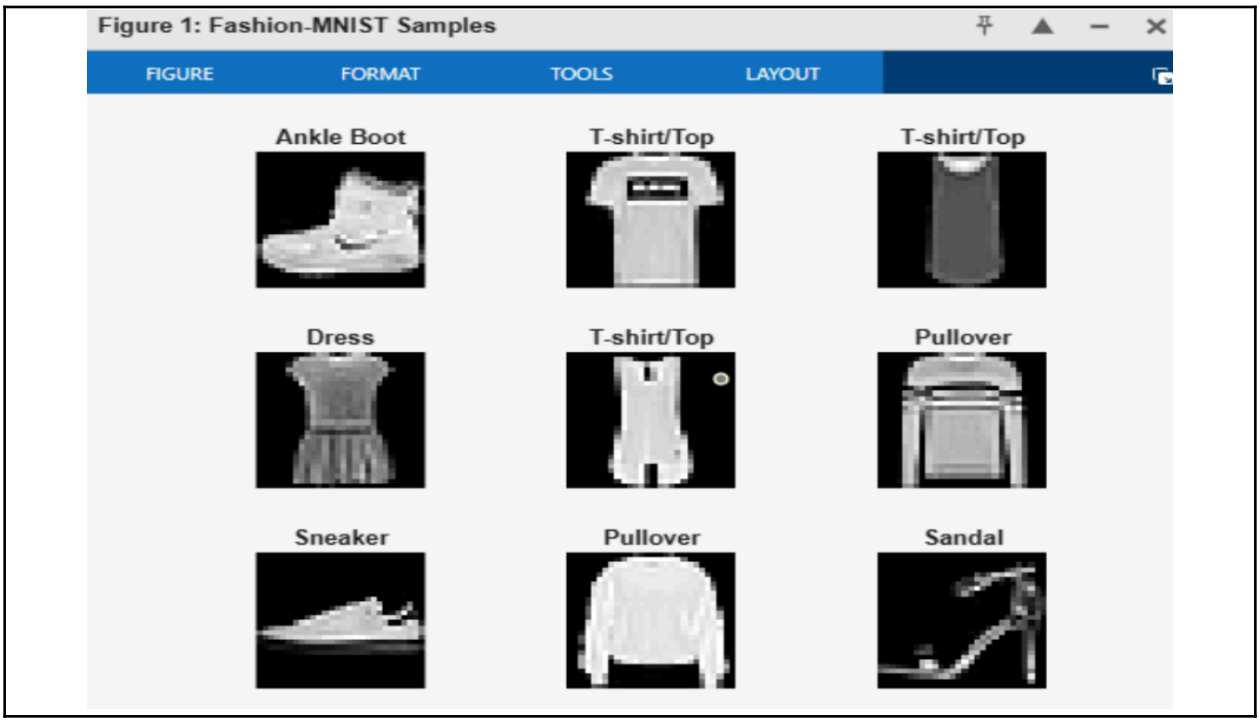


Figure 1: Fashion-MNIST Samples
FIGURE
FORMAT
TOOLS
LAYOUT
Ankle Boot
T-shirt/Top
T-shirt/Top
Dress
T-shirt/Top
Pullover
Sneaker
Pullover
Sandal

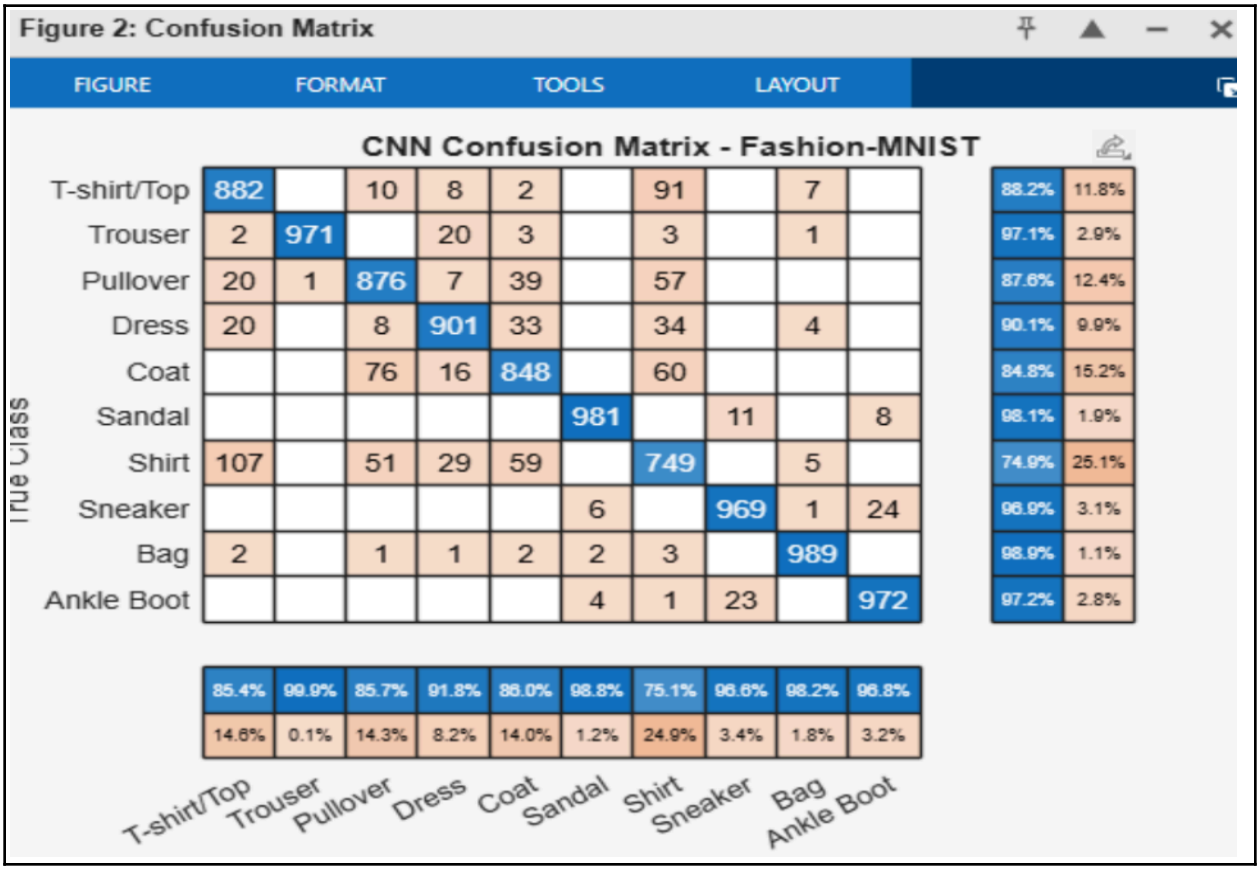


Figure 2: Confusion Matrix
FIGURE
FORMAT
TOOLS
LAYOUT
CNN Confusion Matrix - Fashion-MNIST
True Class
T-shirt/Top 882 10 8 2 91 7 88.2% 11.8%
Trouser 2 971 20 3 3 1 97.1% 2.9%
Pullover 20 1 876 7 39 57 87.6% 12.4%
Dress 20 8 901 33 34 4 90.1% 9.9%
Coat 76 16 848 60 84.8% 15.2%
Sandal 981 11 8 98.1% 1.9%
Shirt 107 51 29 59 749 5 74.9% 25.1%
Sneaker 6 969 1 24 96.9% 3.1%
Bag 2 1 1 2 2 3 989 98.9% 1.1%
Ankle Boot 4 1 23 972 97.2% 2.8%
85.4% 99.9% 85.7% 91.8% 86.0% 98.8% 75.1% 96.6% 98.2% 96.8%
14.6% 0.1% 14.3% 8.2% 14.0% 1.2% 24.9% 3.4% 1.8% 3.2%
T-shirt/Top
Trouser
Pullover
Dress
Coat
Sandal
Shirt
Sneaker
Bag
Ankle Boot

**Conclusion**

CNNs are powerful deep learning models for image processing tasks. Using Keras and TensorFlow, designing and training CNNs becomes efficient and accessible. Datasets like MNIST and CIFAR are ideal for understanding CNN concepts and building real-world applications.